Robert Jankowski

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Academic Appointments

Postdoctoral Researcher Nov 2026 – present
Indiana University Bloomington
Advisor: Maksim Kitsak

Postdoctoral Researcher Nov 2025–Oct 2026
TU Delft
Topic: Shortest paths in large incomplete networks
Advisor: Maksim Kitsak

Education

Ph.D. in Physics 2021–2025
University of Barcelona
Thesis: *Hyperbolic Cartography of Complex Networks*
Supervisors: M. Ángeles Serrano and Marián Boguñá

Visiting Researcher, Luddy Center for Artificial Intelligence, Indiana University, Aug–Dec 2024
Hosts: Santo Fortunato and Filippo Radicchi

M.Sc. in Applied Physics 2020–2021
Warsaw University of Technology
Thesis: *Interactions in signed complex networks*
Supervisor: Piotr Górski

B.Sc. in Applied Physics 2016–2020
Warsaw University of Technology
Thesis: *Predicting election results from opinion polls using machine-learning methods*
Supervisor: Julian Sienkiewicz

Publications

Peer-reviewed

R. Jankowski, M. Kitsak, D. Celińska-Kopczyńska. *Hyperbolic Graph Embedders for Link Prediction and Topology Reconstruction.* The Fifth Learning on Graphs Conference (accepted), (2026)

L. B. Rosselló, **R. Jankowski**, H. Bersini, M. Bogunã, M. Á. Serrano. *Trade-offs between structural richness and perceptual robustness in music network representations.* Phys. Rev. Research, **8**, 033127 (2026)

R. Aliakbarisani*, **R. Jankowski***, M. Á. Serrano, M. Bogunã. *HypBench: Hyperbolic Benchmark for Graph Neural Network Performance.* IEEE Transactions on Neural Networks and Learning Systems (2026)

A. F. Marcús, **R. Jankowski**, M. V. Miñana, C. Casacuberta, M. Á. Serrano. *Chordless cycle filtrations for dimensionality detection in complex networks via topological data analysis.* Nature Communications, **17**, 6105 (2026)

R. Jankowski, F. Radicchi, M. Á. Serrano, M. Bogunã, S. Fortunato. *Task complexity shapes internal representations and robustness in neural networks.* Machine Learning: Science and Technology, **7**, 025045 (2026)

G. Bernárdez, L. Telyatnikov, M. Papillon, ..., A. Weers, **R. Jankowski**, L. Cino, ..., N. Miolane. *Topological Deep Learning Challenge 2025: Expanding the Data Landscape.* Proceedings of the 1st Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS 2025), Proceedings of Machine Learning Research **321**, 4–14 (2026)

M. Kim, L. Cirigliano, C. Castellano, H. Sun, **R. Jankowski**, A. Poggialini, F. Radicchi. *Shortest-path percolation on scale-free networks.* Physical Review E, **113**, 014314 (2026)

R. Jankowski, R. Aliakbarisani, M. Á. Serrano, M. Bogunã. *Mapping bipartite networks into multidimensional hyperbolic spaces.* Communications Physics **9**, 35 (2026)

R. Jankowski, P. Hozhabrierdi, M. Bogunã, M. Á. Serrano. *Feature-aware ultra-low dimensional reduction of real networks.* npj Complexity **1**, 13 (2024)

R. Jankowski, A. Allard, M. Bogunã, M. Á. Serrano. *The D-Mercator method for the multidimensional hyperbolic embedding of real networks.* Nature Communications **14**, 7585 (2023)

R. Jankowski, A. Chmiel. *Role of Time Scales in the Coupled Epidemic-Opinion Dynamics on Multiplex Networks.* Entropy **24**, 105 (2022)

R. Jankowski, J. Sienkiewicz. *Determining Crucial Factors for the Popularity of Scientific Articles.* Acta Physica Polonica A **138**, 41–47 (2020)

*Equal contribution.

Preprints / Under Review

B. L. Nortier, J. Kolk, **R. Jankowski**, S. Dobson, M. Á. Serrano, F. Battiston, M. Boguñá. *Latent geometry organizes higher-order interactions.* arXiv:2609.07906

R. Jankowski, P. Almagro-Blanco, M. Boguñá, M. Weber, M. Á. Serrano. *Train Small, Deploy Large: Zero-Shot GNN Transfer Through Geometric Renormalization.* arXiv:2607.27767

M. V. Miñana, **R. Jankowski**, A. F. Marcús, R. Ballester, M. Á. Serrano, C. Casacuberta. *Motif-based filtrations for persistent homology: A framework for graph isomorphism and property prediction.* arXiv:2604.15265.

Grants and Awards

Grants

External collaborator, YOUNG PW II grant "Application of the low-dimensional representation of real networks for localization of information source" (with Robert Paluch), Warsaw University of Technology, 2024–2026 (150,000 PLN).

Bridge grant with Lluc Bono Rosselló, Young Researchers of the Complex Systems Society (yrCSS), 2024.

Individual Funding

AccelNet-MultiNet program, 2024.

Predoctoral grant FI-SDUR, 2022–2025.

SECS scholarship (yrCSS), 2022.

Awards

Best Contributed Talk, Network Science Informs AI Satellite at NetSci, 2026.

Best Student Presentation, SFINKS Conference, 2021.

Best engineering thesis in Poland, Section Physics in Economy and Social Sciences of Polish Physical Society, 2021.

Rector's scholarship for academic performance: 2020 (3rd/317), 2019 (top 5), 2018.

Talks and Posters

Invited Talks

Understanding Neural Networks with Network Science.

Network Science Institute, Northeastern University London, London, 2026

Network geometry and multidimensional hyperbolic maps of real networks.

Faculty of Physics, Warsaw University of Technology, Poland, 2024

Contributed Talks

Task complexity shapes internal representations and robustness in neural networks.

Network Science Informs AI (NSIA) Satellite at NetSci 2026, Boston, USA, 2026

IAIFI PhD Summer School, Boston, USA, 2025

Network representations reveal structured uncertainty in music.

XIII Polish Symposium on Physics in Economics and Social Sciences, Warsaw, Poland, 2025

Mapping bipartite networks into multidimensional hyperbolic spaces.

NetSci 2025, Maastricht, The Netherlands, 2025

Feature-aware ultra-low dimensional reduction of real networks.

NetSci 2024, Quebec City, Canada, 2024

The D-Mercator method for the multidimensional hyperbolic embedding of real networks.

APS March Meeting, Minneapolis, USA, 2024

2nd Meeting of the Spanish Chapter of the CSS, Barcelona, Spain, 2024

Statphys28, Tokyo, Japan, 2023

NetSci 2023, Vienna, Austria, 2023

CCS 2022, Palma de Mallorca, Spain, 2022

Role of time scales in coupled epidemic-opinion dynamics on multiplex networks.

NetSci-X 2022, virtual/Porto, Portugal, 2022

The influence of relations in forming interactions among communities on social websites.

SFINKS conference, 2021. Best student presentation

Crucial factors determining the popularity of scientific articles.

10th Polish Symposium on Physics in Economy and Social Sciences, 2019

Posters

Benchmarking Hyperbolic Graph Embeddings for Link Prediction and Topology Reconstruction.

NetSci 2026, Boston, USA, 2026

HypBench: Hyperbolic Benchmark for Graph Neural Network Performance.

Non-Euclidean Foundation Models and Geometric Learning Workshop @ NeurIPS 2025, San Diego, USA, 2025

Task complexity shapes internal representations and robustness in neural networks.

IAIFI Summer Workshop, Boston, USA, 2025

Feature-aware ultra-low dimensional reduction of real networks.

3rd Meeting of the Spanish Society of Complex Systems, Madrid, Spain, 2025

D-Mercator: multidimensional hyperbolic embedding of real networks.

XXIV Congreso de Física Estadística, Pamplona, Spain, 2023

Generating interactions for friendship networks.

NetSci 2023, Vienna, Austria, 2023

From relation to interactions: a case study in Reddit website.

11th Polish Symposium on Physics in Economy and Social Sciences, 2021

Teaching and Outreach

M.Sc. thesis supervisor, TU Delft (2026). Topic: *Shortest Paths and Hyperbolic Geometry.*

Teaching Fellow, DIS Master Class: Network Renormalization, CSH Vienna, May 2025.

Delivered the workshop "Les xarxes complexes com a eina per resoldre problemes" at the University of Barcelona's Festa de la Ciència (2023 and 2024).

Professional Service

Chair of Digital Infrastructure, Network Science Society (2026–present).

Invited Editor, Special Collection on "Network Science for Understanding AI," *Physical Review E.*

Journal Reviewer: *Physical Review E*, *Communications Physics*, *Chaos*, *Scientific Reports*, *PLOS ONE*, *EPJ Data Science*, *Physica A*, *Proceedings of the Royal Society A*.

Program Committee Member: NetSci (2024–2026), CompleNet (2026), ACM KDD (2024), MISDOOM (2022).

Conference and Workshop Organization

Co-organizer, Network Geometry: Theory and Applications, satellite at NetSci 2026, Boston, MA, USA.

Co-organizer, Network Geometry: Theory and Applications, satellite at NetSci 2025, Maastricht, Netherlands.

Co-organizer, Network Geometry: Theory and Applications, satellite at NetSci 2024, Quebec, Canada.

Summer Schools and Workshops

IAIFI PhD Summer School & Summer Workshop Aug 2025
Institute for AI and Fundamental Interactions, Harvard University, USA

Complementarity in Complex Networks Workshop May 2025
Delft, The Netherlands

LOGML Summer School Jul 2024
London, United Kingdom

Visualizing Complexity Science Workshop Aug 2023
Complexity Science Hub, Vienna, Austria.

XI GEFENOL Summer School on Statistical Physics of Complex Systems Jul 2023
Barcelona, Spain

Complex Networks: Theory, Methods, and Applications Summer School May 2022
Lake Como School of Advanced Studies, Italy

Additional Experience

Research Intern Jul–Aug 2021
Samsung R&D Institute Poland.
Developed and evaluated automatic speech recognition systems as part of the Bixby team.

Student Researcher May–Dec 2021
Warsaw University of Technology. Group of Anna Chmiel.
Project "Modeling epidemic spread using comorbidities and social attitudes".

Student Researcher 2020–2021
Warsaw University of Technology. Group of Janusz Hołyst.
Project ALPHORN on signed relations and structural balance (with ETH Zürich).

Research Intern Jul–Sep 2020
Samsung R&D Institute Poland.
Developed and tested deep-learning models for named entity recognition.

Scala Intern & Junior Scala Developer Jul–Nov 2019
TouK, Warsaw
Technologies: *Scala, Akka, Python, Docker, RabbitMQ.*

Software Engineering Intern Jul–Aug 2018
Ilabo, Warsaw.
Developed web applications in C# and .NET Core and performed data analysis in Python.

Art Exhibitions

Capillary Network. Ars Electronica Festival 2025, Linz, Austria.

Capillary Network. Altering Nature: Exploring Life in Computational Art, HKUST (Guangzhou), China, April 2025.